STM32F407 Mixed-Signal Hardware Design

Design Overview & PCB Documentation

Complex Mixed-Signal Board Design Course
Ethernet PHY · STM32F407 · STM32F103 · CH340C · DAC/MIC · 24-bit ADC · 36 W Motor Drivers

Document	MSD-001	**Revision**	V1.0
Designer	Maulik Italiya	**Date**	06-10-2026
Design Tool	Altium Designer	**Main MCU**	STM32F407VGT6

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1 Project Overview

This project is a custom **4-layer mixed-signal embedded control board** built around the **STM32F407VGT6** microcontroller. It was designed as part of a complex mixed-signal board design course and combines digital communication, motor control, Ethernet, USB/UART debugging, high-resolution analog acquisition, stereo audio, and microphone input on a single board powered from a 12 V supply.

The 11-page schematic is organized into dedicated sheets for the STM32F407 controller, Ethernet PHY, two DRV8701 motor-driver channels, 24-bit ADC, stereo DAC and microphone, USB/UART interfaces, the STM32F103 debugger, and power regulation.

1.1 Key Features

- **Controller:** STM32F407VGT6 with 8 MHz crystal and SWD/JTAG debug access
- **Ethernet:** DP83826 10/100 PHY over MII with RJ45 connector and ESD protection
- **Motor control:** 2× DRV8701 gate drivers with external N-MOSFET H-bridges, 12 V / 3 A per channel (36 W each)
- **Precision measurement:** ADS122C04 24-bit $\Delta\Sigma$ ADC for load-cell input over I^2C
- **Audio:** CS43L22 stereo DAC (I^2S) with headphone/line and speaker outputs, plus IMP34DT05 digital microphone
- **Debug & USB:** on-board STM32F103C8T6 JTAG/SWD programmer and CH340C USB-to-UART bridge
- **External interfaces:** UART and CAN logic-level headers
- **Power:** 12 V input with +3.3 V and +1.8 V regulated rails
- **PCB:** 4-layer stack-up with solid GND plane and dedicated power plane

2 System Architecture

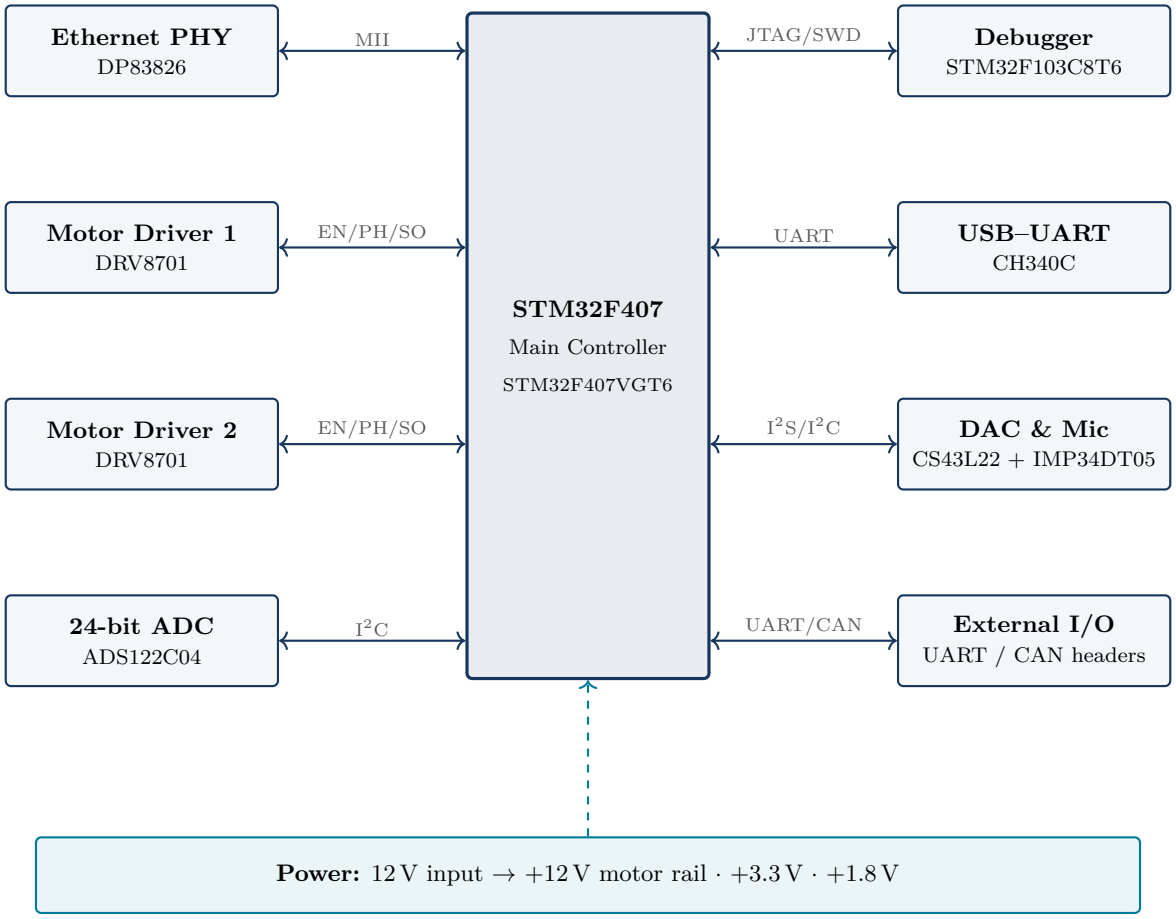


Figure 1. System block diagram (derived from schematic sheet 1).

The STM32F407 is the central controller. It talks to the Ethernet PHY over MII, to the two DRV8701 motor-driver channels through one PWM output, one GPIO and one analog current-sense input each, to the ADS122C04 over I^2C, to the CS43L22 DAC over I^2S (with I^2C for configuration), to the microphone through a clock/data pair, and to the CH340C through UART. The STM32F103 acts as a standalone JTAG/SWD debugger/programmer for the STM32F407.

3 Schematic Sheet Index

Sheet	Title	Content
1	**Schematic Blocks**	System block diagram and interface summary
2	**Power Budget**	12 V input and +3.3 V load budget
3	**STM32F103 Debugger**	STM32F103C8T6 programmer, JTAG/SWD header, USB mini-B
4	**Ethernet PHY**	DP83826, 25 MHz crystal, RJ45, ESD protection, status LEDs
5	**DRV8701 Motor Driver 1**	DRV8701 gate driver, DMHT6016LFJ-13 MOSFET bridge, motor connector
6	**CH340C USB–UART**	CH340C USB-to-UART bridge and USB mini-B
7	**24-bit ADC**	ADS122C04 load-cell front end and connector
8	**Stereo DAC & Mic**	CS43L22 DAC, audio outputs, IMP34DT05 microphone
9	**STM32F407 Controller**	MCU, power decoupling, clock, reset, UART/CAN headers
10	**DRV8701 Motor Driver 2**	Second motor channel (same topology as sheet 5)
11	**Power Supply**	12 V input connector, +3.3 V and +1.8 V regulators

4 Subsystem Design

4.1 Main Controller (Sheet 9)

Item	Implementation
Device	STM32F407VGT6 (LQFP-100)
Clock	8 MHz crystal (X2) with 30 pF load capacitors
Reset	Tactile switch (SW3) with 0.1 µF capacitor on NRST
Supply	+3.3 V; 100 nF decoupling per VDD pin, 4.7 µF bulk, 2.2 µF on VCAP1/VCAP2
Analog supply	VDDA / VREF+ fed through ferrite bead FB4 with local capacitors
Debug	JTAG/SWD signals from the STM32F103 debugger (sheet 3)
External I/O	J8 (UART) and J9 (CAN logic signals), 4-pin XH connectors

4.2 STM32F103 Debugger / Programmer (Sheet 3)

Item	Implementation
Device	STM32F103C8T6
Clock	8 MHz crystal (X1) with 30 pF load capacitors
USB	USB 2.0 mini-B (J2) with 1.5 kΩ D+ pull-up
Debug header	6-pin 2.54 mm header (J1) for the debugger's own JTAG/SWD
Target link	JTMS, JTCLK and NTRST lines routed to the STM32F407
Reset	Tactile switch (SW1) with 10 kΩ pull-up

4.3 Ethernet (Sheet 4)

Item	Implementation
PHY	DP83826IRHBR, MII interface to the STM32F407
Clock	25 MHz crystal (Y1) with 20 pF load capacitors
Bias	6.49 kΩ RBIAS resistor
Supply filtering	Ferrite bead FB1 with 10 µF / 1 µF / 0.1 µF / 10 nF decoupling on the analog supply
Connector	RJ45 (J3) with four ESD3V3D9-TP protection diodes on the signal pairs
Chassis	EARTH connection through 1 MΩ in parallel with 10 nF
Status	Collision (COL) and carrier-sense (CRS) red LEDs; PHY reset switch (SW2) plus MCU-driven PHY_RESET

4.4 Motor Control (Sheets 5 and 10)

Two identical channels, each with a gate driver, an external N-MOSFET bridge, a low-side current-sense resistor and a motor connector.

Item	Implementation
Gate driver	DRV8701ERGET (U3, U9), controlled with EN (PWM) and PH (direction)
Power stage	DMHT6016LFJ-13 N-MOSFET bridge (Q1, Q2)
Current sense	50 mΩ shunt (R20, R44); SO output to an MCU analog input
Fault reporting	nFAULT and SNSOUT signals to the MCU
Supply	+12 V with 100 µF bulk, 1 µF and 0.1 µF decoupling per channel
Connector	2-pin B2B-XH-A (J4, J10)
Per-channel interface	1 GPIO, 1 PWM, 1 analog input
Power rating	12 V × 3 A = 36 W per channel

4.5 24-bit Analog Acquisition (Sheet 7)

Item	Implementation
Device	ADS122C04IPW, 24-bit ADC with I^2C interface
I^2C address	Set by A0/A1 tied to GND (schematic note: `10000000`)
Control lines	RESET and DRDY connected to the MCU
Input filtering	1 kΩ series resistors, 10 nF to ground on each input, 0.1 µF differential capacitor
Supply filtering	Ferrite bead FB2 with 0.1 µF decoupling on AVDD
Connector	J6, 4-pin 1.25 mm: AIN0, AIN1, +3.3 V, GND (load cell)

4.6 Audio and Microphone (Sheet 8)

Item	Implementation
DAC	CS43L22-CNZ stereo DAC with headphone/speaker amplifiers
Digital audio	I^2S: MCLK, SCLK, LRCK, SDIN from the STM32F407
Control	I^2C (SCL/SDA shared with the ADC bus), RESET_DAC from the MCU; schematic address note `10010100`
Supplies	+3.3 V analog/headphone/logic rails and +1.8 V digital core, with local decoupling
Outputs	4-pin stereo jack (J7) with output filter network; 4-pin header (J12) for OUTA± / OUTB±
Microphone	IMP34DT05TR digital MEMS microphone, clock (CLK_MIC) and data (DOUT_MIC) to the MCU

4.7 USB-to-UART (Sheet 6)

Item	Implementation
Device	CH340C USB-to-UART converter
Interface	TXD and RXD to the STM32F407 UART; RTS# routed to the MCU as CH340_RST
Connector	USB 2.0 mini-B (J5)
Supply	+3.3 V with 0.1 µF decoupling (V3 pin tied to VCC)

5 Power Architecture

The board is supplied from a single 12 V input (3-pin 2.5 mm terminal block, J11) with three 100 µF bulk capacitors. The +12 V rail feeds both motor-driver channels directly. Each low-voltage regulator is fed through a MBRS130LT3G Schottky diode (D5, D7) and uses an LD1117-series LDO with input and output decoupling. Zener diodes (D6,

D8) are fitted on the regulator outputs. Four mounting holes are tied to EARTH.

5.1 Power Budget (Sheet 2)

Load	Current	Rail
Motor Driver 1	3 A	+12 V
Motor Driver 2	3 A	+12 V
STM32F407	240 mA	+3.3 V
STM32F103	150 mA	+3.3 V
Ethernet PHY	55 mA	+3.3 V
ADC IC	10 mA	+3.3 V
DAC	10 mA	+3.3 V
Microphone chip	10 mA	+3.3 V
+3.3 V subtotal	**475 mA**	+3.3 V
12 V total (6.13 A + 25 % margin)	**7.66 A**	+12 V input

Note: The values above are the design power budget from the schematic. Final current capability and thermal performance should be verified on the assembled PCB under measured operating conditions.

6 4-Layer PCB Stack-Up

The PCB uses a **4-layer, approximately 1.6 mm thick stack-up** with a dedicated ground plane and a dedicated power plane.

Layer	Function	Purpose
L1	Top signal + components	Primary component placement and signal routing
L2	Solid GND plane	Continuous ground reference and low-impedance return-current path
L3	Dedicated power plane	Power distribution only; segmented for +12 V, +3.3 V and +1.8 V
L4	Bottom signal + components	Secondary signal routing and backside component placement

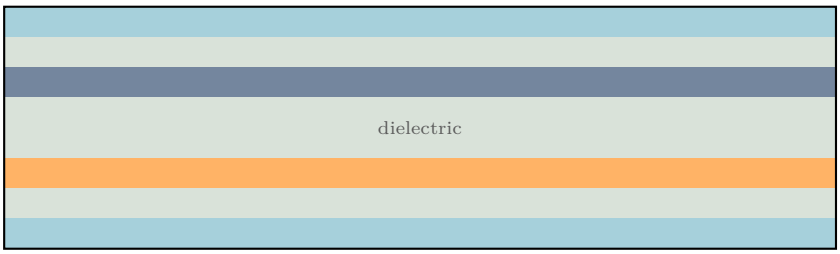


Figure 2. Conceptual stack-up (not to scale; approx. 1.6 mm total).

The second layer is kept as a **solid GND plane** to provide a continuous reference plane and controlled return-current paths. The third layer is used exclusively for power distribution, with separate copper regions for **+12 V**, **+3.3 V** and **+1.8 V**. No signal routing is placed on L3.

Stack-up note: The nominal 1.6 mm board thickness is the design configuration. Exact dielectric thicknesses and copper weights depend on the selected fabrication process and manufacturer.

7 Mixed-Signal Layout Strategy

The PCB combines high-current motor-control electronics with sensitive analog, audio and high-speed digital circuitry. The main layout rules are:

- Keep high-current motor paths and switching nodes separated from sensitive analog and audio circuitry.
- Use the solid L2 ground plane as the primary return path and reference plane.
- Use the dedicated L3 plane to distribute +12 V, +3.3 V and +1.8 V without signal routing.
- Place decoupling capacitors close to IC supply pins to minimize high-frequency supply-loop areas.
- Keep Ethernet MII/PHY routing controlled and physically separated from noisy motor-switching regions.
- Protect and filter the load-cell/ADC inputs, since these signals are sensitive to noise coupling.
- Keep the audio DAC and microphone circuitry away from high-current motor switching and noisy power paths.

8 PCB Design Overview

The completed PCB places components and routing on both the top and bottom sides. The STM32F407 section is located near the centre of the board, with the communication, power, motor-control and analog sections distributed around it.

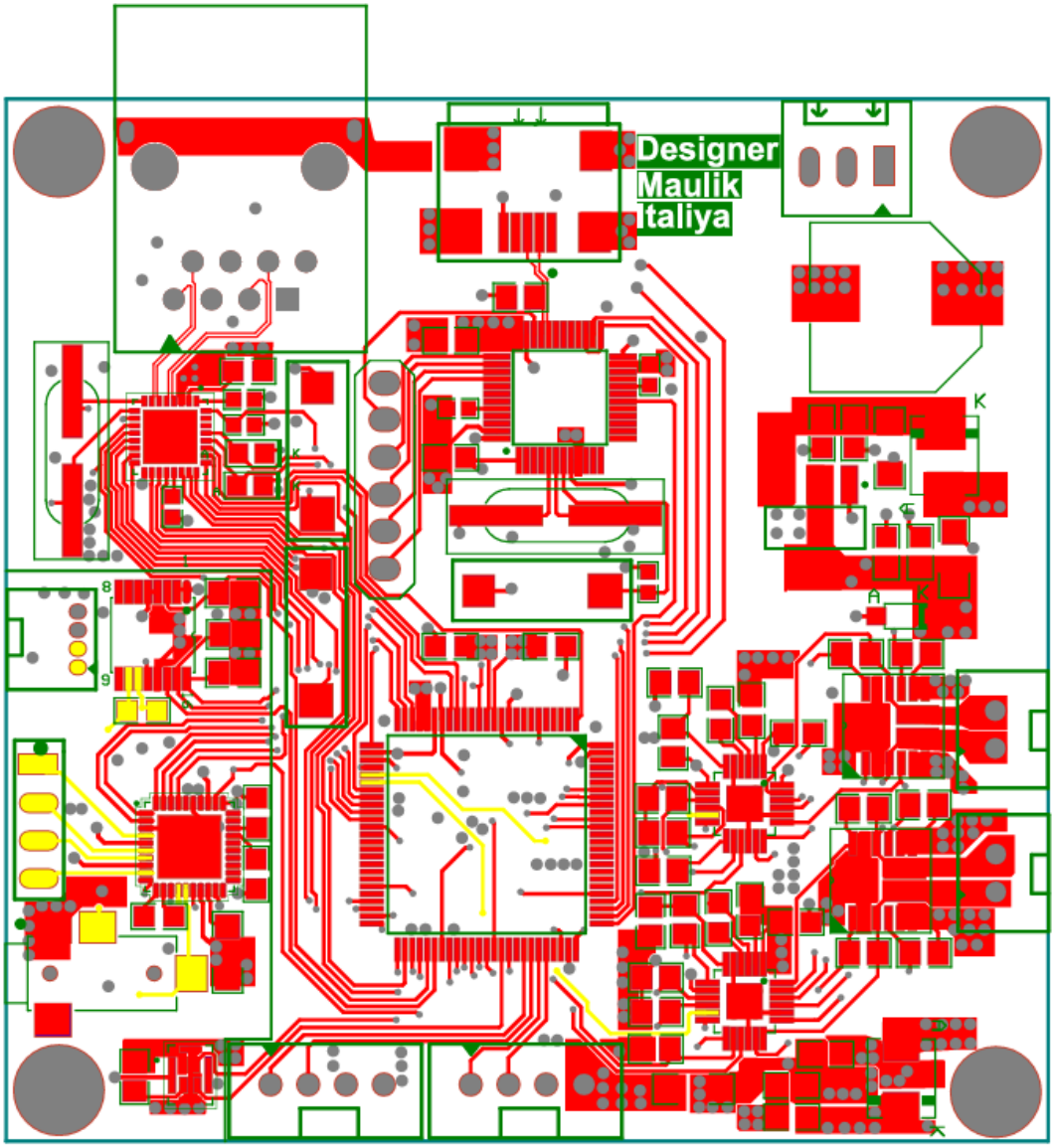


Figure 3. PCB top layer (L1).

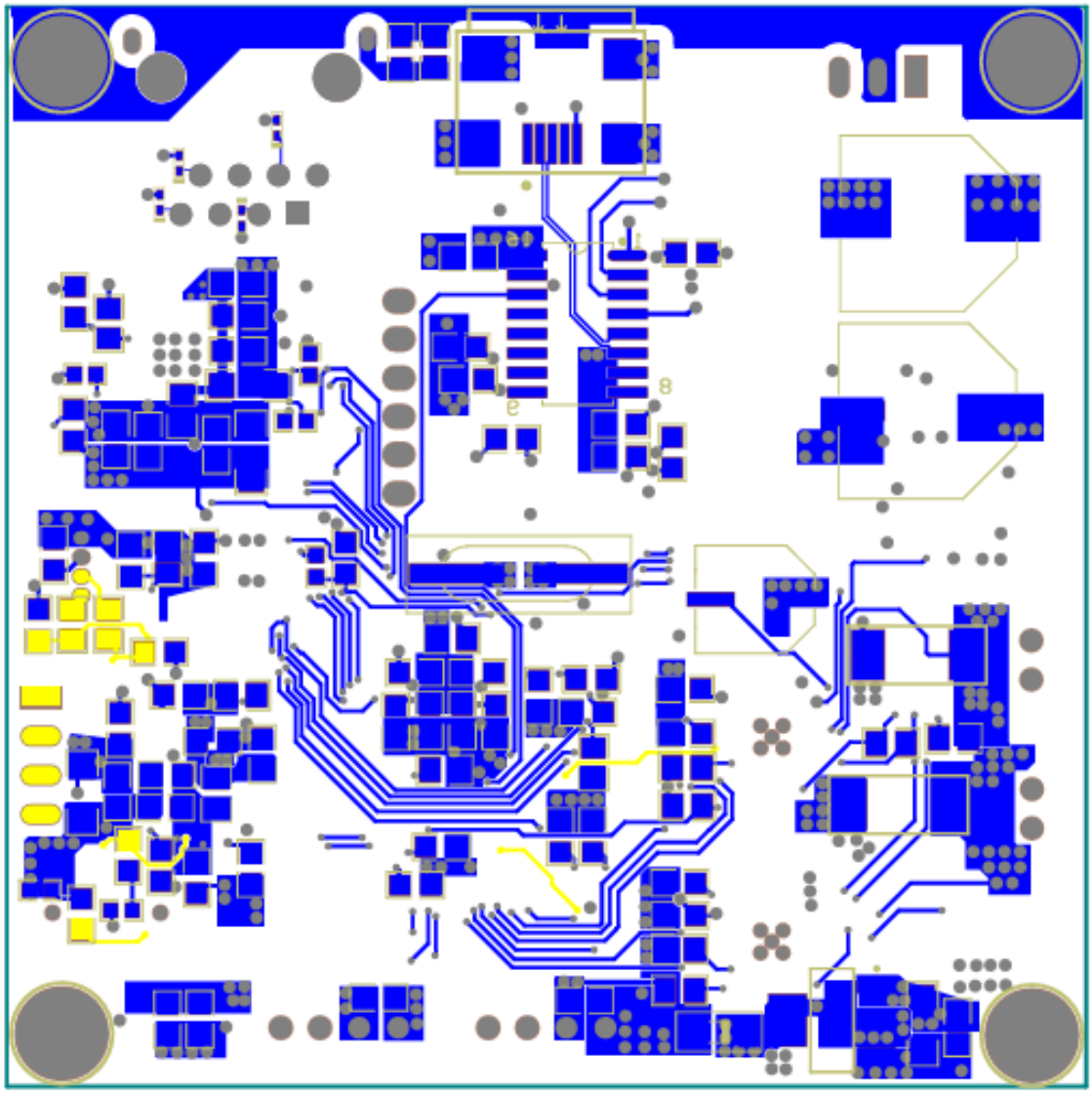


Figure 4. PCB bottom layer (L4).

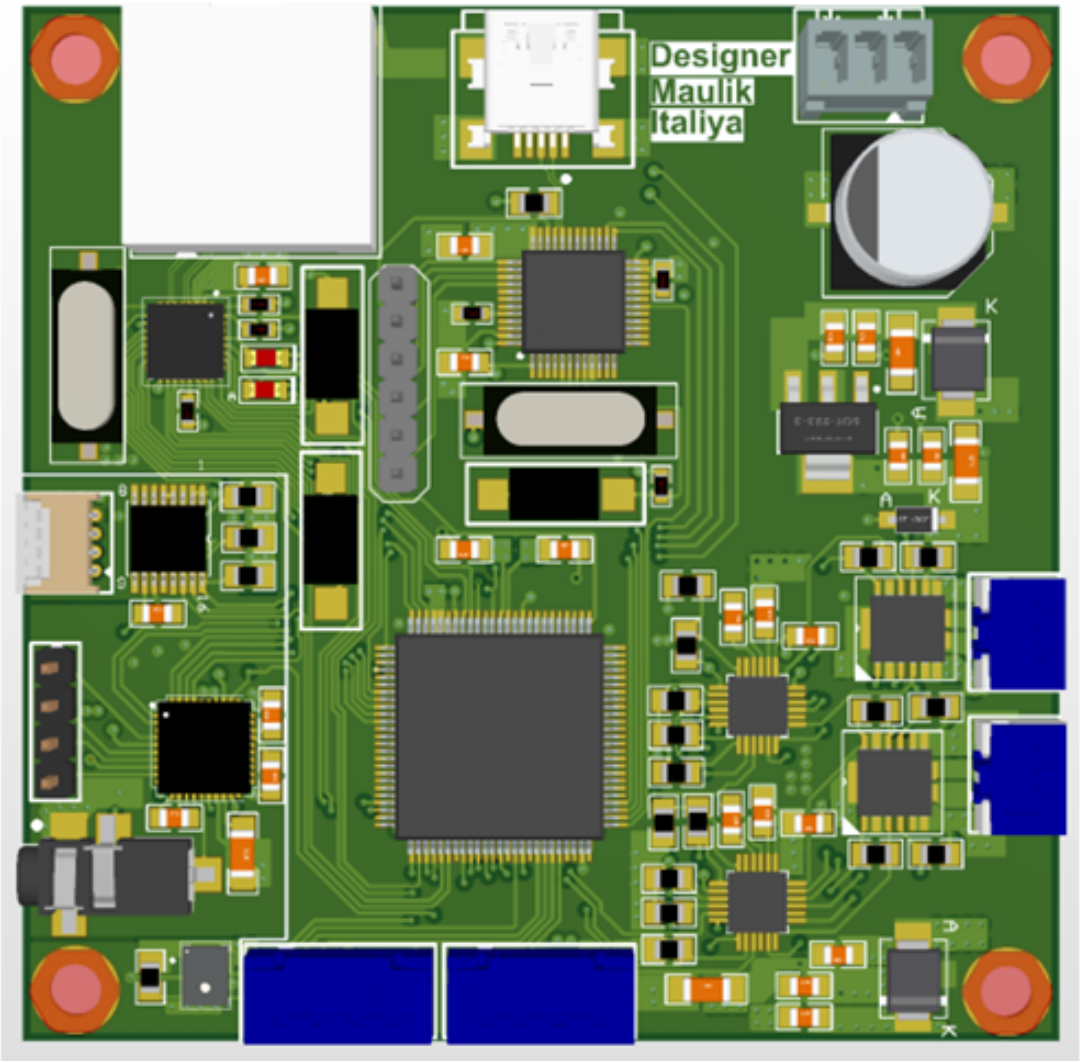


Figure 5. PCB 3D view – top.

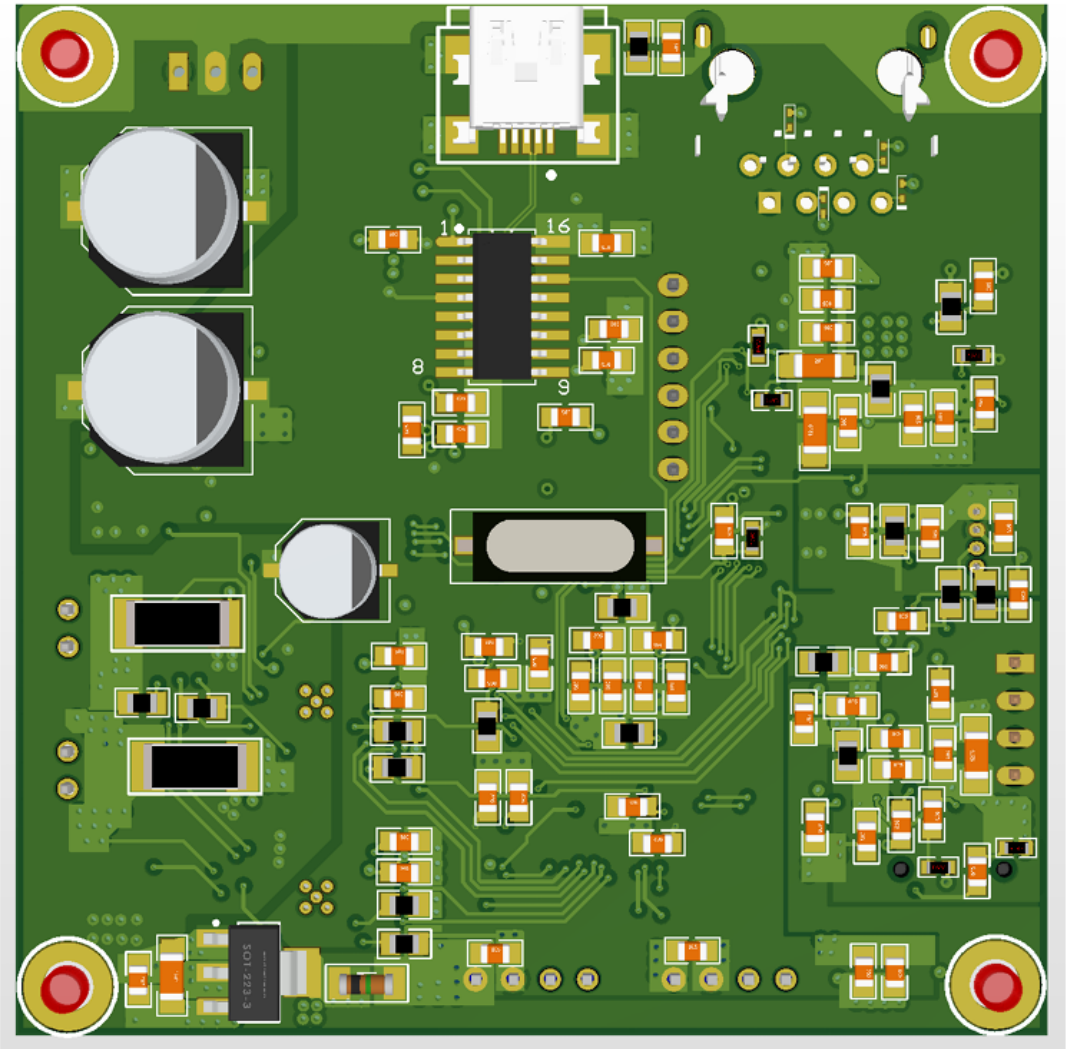


Figure 6. PCB 3D view – bottom.

9 Project Documentation

- 11-page electrical schematic (PDF)
- Top-layer and bottom-layer PCB documentation
- 4-layer PCB stack-up information
- PCB 3D views
- Design overview and architecture (this document)
- Bill of Materials
- Manufacturing documentation (Gerber / drill files)

10 Design Summary

Document	MSD-001
Revision	V1.0
Date	06-10-2026
Main MCU	STM32F407VGT6
Board type	4-layer mixed-signal embedded control board
PCB thickness	Approximately 1.6 mm
Layer 2	Solid GND plane
Layer 3	Dedicated +12 V / +3.3 V / +1.8 V power plane
Design tool	Altium Designer
Designer	Maulik Italiya

10.1 Revision History

Revision	Date	Description
V1.0	**06-10-2026**	Initial release: schematic, PCB layout and design overview